DEWDRIVE · PRODUCT REQUIREMENTS AND SPECIFICATIONS

# DewDrive product requirements and specifications

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## DOCUMENT CONTROL

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| Document ID | DWD-PRS-001 | Version | 1.1 |
| Status | Draft | Date | 2026-10-25 |
| Author | Amish Chadha | License | CERN-OHL-S-2.0 |
| Project TRL | TRL 3 Proof of concept | | |
| Repository | github.com/BoujeeEnjinia1701/dewdrive | | |
| Source | docs/03-requirements.md @ 4f86e9e | | |

## REVISION HISTORY

| Version | Date | Author | Change |
|---|---|---|---|
| 1.1 | 2026-10-25 | Amish Chadha | Amish's decisions of 2026-10-25: PR-010 design wind a 25 m/s gust (item 4); fan cut-out 65 % RH in the assumptions (item 3) |
| 1.0 | 2026-10-25 | Amish Chadha | Kit 1.8 pipeline: DWD-REQ-001 migrated to DWD-PRS-001 (each R becomes a PR with the old number under Was, tolerance, units and the user requirement it traces to); PR-014 added for UR-009; status from DWD-CAL-001 v0.8 after the STEM check fixes and independent review (D16 to D20); PR-001 not met |
| 0.9 | 2026-10-02 | Amish Chadha | Status from DWD-CAL-001 v0.7 (script re-run; R9 met as restated); R11 over the target by $119 after the 130 °C glazing and the label |
| 0.8 | 2026-10-02 | Amish Chadha | R9 restated by Amish on 2026-10-02 for a lift with the trays and deck taken out, with an empty-before-lifting label (DWD-DEC-001, item 2); R9 met |

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| 0.7 | 2026-10-01 | Amish Chadha | Budget treated as a value-engineering target |
| 0.6 | 2026-10-01 | Amish Chadha | Status from DWD-CAL-001 v0.4 for the constructable design (DWD-DDR-003); R9 and R11 not met, open for Amish |
| 0.5 | 2026-09-26 | Amish Chadha | Budget top-up approved by Amish (R11 target $520; status from DWD-CAL-001 v0.3) |
| 0.4 | 2026-09-25 | Amish Chadha | Recommendations accepted by Amish (DDR-002); R7 restated for the drip screens; R8 and R10 wording; status from DWD-CAL-001 v0.2 |
| 0.3 | 2026-09-25 | Amish Chadha | TRL 3 update (design point and budget decided by Amish in DWD-DDR-001; R11 redefined to $500; status column from DWD-CAL-001) |
| 0.2 | 2026-09-25 | Amish Chadha | First measurable requirements for TRL 2, with a design point and concept status |
| 0.1 | 2026-09-24 | Amish Chadha | Initial scaffold |

# DewDrive product requirements and specifications

What the design must do, each requirement traced to a user need in DWD-URS-001. This document replaces DWD-REQ-001 (v0.9); the old R number is kept under "Was". The requirements were checked by calculation at TRL 3 in DWD-CAL-001 v0.8, for the design decided by Amish on 2026-09-25 (DWD-DDR-002), made buildable in DWD-DDR-003 and corrected by the STEM check and its independent review on 2026-10-25 (DWD-DEC-001, D16 to D20): black absorber plates on the beds, PTFE-coated brine-wetted parts, an open tray floor of mesh on bars, a painted condenser underside and stronger ground anchors. On paper the design meets 10 of 14 requirements. **PR-001 is not met:** the design collects about 0.44 L per day at the design point against 0.5 L, and 0.37 to 0.41 L under the calculation's sensitivities (C9, C12, C13); the ways to recover it are for Amish (DWD-DEC-001, open item 2). PR-011 (cost) is over its value-engineering target, at an estimated $695 against $520. PR-006 and PR-010 can only be verified by test, and PR-005 is at risk if the fins shed less heat than the correlation says. Targets are proposals for review with a co-design partner, not user-validated needs. Method: STANDARDS section 22.

## Design point

Decided by Amish on 2026-09-25 (DWD-DDR-001, D8). Unless a requirement says otherwise, it applies at this design point:

- Night: 10 h adsorption at 20 °C air and 40 % RH (6.92 g/m³ of vapour).
- Day: 6.0 kWh/m² of sun on the tilted aperture, of which 4.24 kWh/m² falls in the 09:00 to 15:00 window (DWD-CAL-001, C1); air at 35 °C.
- Low-humidity case: night air at 20 °C and 25 % RH, same sun.

## Requirements

Table 1. Product requirements and specifications. Values and tags are from DWD-CAL-001 v0.8.

| ID | Traces to | Category | Title | Requirement | Target | Tolerance | Units | Verification | Status | Wa |
|---|---|---|---|---|---|---|---|---|---|---|
| PR-001 | UR-001 | Performance | Water at the design point | The unit shall collect water from night air at the design point | 0.5 | min | L/day | Analysis (DWD-CAL-001 cyclic balance), TRL 3; field test at TRL 4 | Not met: 0.44 (C3); 0.37 to 0.41 under the sensitivities (C9, C12, C13) | R1 |
| PR-002 | UR-002, UR-001 | Performance | Water in dry nights | The unit shall collect water when the night air is at 25 % RH | 0.25 | min | L/day | Analysis (DWD-CAL-001), TRL 3; field test at TRL 4 | Met on paper: 0.31 (C4) | R2 |
| PR-003 | UR-008 | Performance | Sun-only regeneration | The bed shall be regenerated by the sun alone and reach the target temperature by midday on a clear day at 35 °C | 75 | min | °C | Analysis (collector balance), TRL 3; logged at TRL 4 | Met on paper: 105 °C at noon (C2) | R3 |
| PR-004 | UR-008 | Electrical | Own electrical loads | The fan and logger shall use no more than the target energy per day, supplied by the on-board PV panel, with at least 2 days of battery autonomy | 30 | max | Wh/day | Analysis (energy budget), TRL 3; measurement at TRL 4 | Met on paper: 23.6 against 35 Wh/day; 2.6 days (F1, F2) | R4 |
| PR-005 | UR-001 | Performance | Condenser rise | The condenser shall stay within the target rise above ambient at the peak of desorption | 15 | max | K | Analysis (condenser balance, painted underside D18), TRL 3; logged at TRL 4 | At risk: 11.9 K on the correlation (C2); about 14 to 16 K if the fins under the floor shed less heat (independent review) | R5 |
| PR-006 | UR-004 | Materials | Water quality | Every wetted part shall be food-grade, and chloride in the | 250 | max | mg/L | Inspection (materials review), TRL 3; laboratory | To verify at TRL 4; materials | R6 |

| ID | Traces to | Category | Title | Requirement | Target | Tolerance | Units | Verification | Status | Wa |
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| | | | | collected water shall stay below the target | | | | water test at TRL 4 | chosen to meet it | |
| PR-007 | UR-012, UR-004 | Safety | Salt containment | No $CaCl_2$ solution shall leave the tray and drip-screen assembly or reach the condenser after a night at 90 % RH; the pores shall stay below the target fill | 100 | max | % of pore volume | Analysis (pore and sump volumes, DWD-CAL-001 E), TRL 3; chamber test at TRL 4 | Met on paper: 47 % after one night, 54 % after three with the fan rule; sumps 0.58 L (E2 to E6) | R7 |
| PR-008 | UR-003 | Usability | Two actions per day | Daily operation shall need no more than two user actions (open the flaps at dusk, close them at dawn) taking the target time | 5 | max | min/ day | Analysis (task analysis), TRL 3; co-design sessions at TRL 4 | Met by design | R8 |
| PR-009 | UR-007 | Packaging and transport | Portable | The box, lifted with the trays and deck taken out, shall weigh no more than the target (two-person lift), carry an "Empty before lifting" label and be separable from the stand; two people shall set it up in 30 min or less. Restated by Amish on 2026-10-02 (DWD-DEC-001, item 2) | 35 | max | kg | Analysis (mass from model components), TRL 3; trial at TRL 4 | Met on paper: 23.4 kg; 36.1 kg with everything inside; 61.1 kg in all (G1, G2) | R9 |
| PR-010 | UR-006 | Physical architecture | Survive the site | The unit shall stay put in the target wind gust with two ground anchors (or ballast), use UV-stable glazing, | 25 | min | m/s | Analysis (wind load), TRL 3; supplier data and cycling test at TRL 4 | To verify at TRL 4: two anchors of 234 N each at 25 m/s, specified 250 N, or 96 | R10 |

| ID | Traces to | Category | Title | Requirement | Target | Tolerance | Units | Verification | Status | Wa |
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| | | | | and keep the sorbent at 80 % or more of its capacity over 300 cycles; the target is a 25 m/s gust until the site value is known from the national wind annex (decided by Amish, 2026-10-25, DWD-DEC-001 item 4; was 20 m/s) | | | | | kg of ballast (H2 to H4) | |
| PR-011 | UR-010 | Physical architecture | Cost | The parts cost shall not exceed the value-engineering target in `project.yaml` (a hypothetical control target, not a limit) | 520 | max | USD | Analysis (priced BOM), TRL 3 | Over target by $175: $695 (J1) | R11 |
| PR-012 | UR-011 | Software | Record performance | The logger shall record air temperature and RH and the bed and condenser temperatures at the target interval for 30 days or more; the daily water mass is recorded by weighing | 5 | max | min | Analysis (storage estimate), TRL 3; demonstration at TRL 4 | Met on paper: 8,640 records, 0.41 MB (I1) | R12 |
| PR-013 | UR-005 | Safety | Protect users | Surfaces a person touches in normal use shall stay at or below the target; the glazing is labelled hot; no exposed conductor above 60 V DC | 60 | max | °C | Analysis (surface temperatures), TRL 3; logged at TRL 4 | Met on paper: glazing outer skin 55 °C, fins 47 °C; 12.8 V DC (D1 to D3) | R13 |

| ID | Traces to | Category | Title | Requirement | Target | Tolerance | Units | Verification | Status | Wa |
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| PR-014 | UR-009 | Physical architecture | Local build and repair | Every made part shall be made with hand tools, a drill, a rivet tool, a small press brake or folding bar and a coating shop, and every bought part shall come from a general supplier; no part needs special tooling | 0 | max | parts needing special tooling | Inspection (build plan DWD-BLD-001 and BOM), TRL 3; build at TRL 4 | Met by design | non |

Categories: physical architecture, performance, materials, usability, packaging and transport, safety, electrical, software. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

## Assumptions

- The composite (25 wt % $CaCl_2$: 1.0 kg in 3.0 kg of mesoporous silica gel) is modelled from bulk salt thermodynamics: at equilibrium it holds 0.397 g/g at 40 % RH and 0.258 g/g at 25 % RH (DWD-CAL-001, A4). The yield is set by how fast the bed takes up and gives off water, not by these capacities. Measured isotherms and uptake rates of the actual composite must replace the model.
- The black absorber plates (DWD-DEC-001, D16) give an effective absorptance of about 0.89; the share of light passing the holes that the white beads and the plate underside absorb (0.70) is an estimate to measure at TRL 4 (DWD-CAL-001, C11).
- The day ends with the salt dried to about the dihydrate; the night starts with the bed at air temperature.
- With the fan stopped above 65 % RH (restarting below 60 %, DWD-DEC-001 item 3), natural airflow through the open flaps is taken as 4 $m^3/h$ (10 % of the fan flow); this is an assumption to check by test.
- The tray floor's open share for the vapour (0.59: a mesh about 60 % open on bars 97.5 % open) is an estimate to confirm from the supplier's mesh data (DWD-DEC-001, D19).
- One unit is a supplement: about 0.44 L per day is about one sixth of one person's survival drinking need of 2.5 to 3 L per day (WHO technical note 9).

> **Safety:** PR-006, PR-007, PR-010 and PR-013 exist because of hazards: untreated water, a corrosive salt solution, wind loads on a 1.1 $m^2$ panel, and hot surfaces. They cannot be waived without a documented decision by Amish.